

The Maximum-Alpha Selection Principle: BST Has Zero Inputs

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Abstract

We prove that the fine structure constant $\alpha(n)$, as derived from the Wyler-BST formula over odd-dimensional type IV symmetric domains D_{IV}^n , achieves its unique global maximum at $n = 5$. The proof proceeds in three steps: (i) an explicit ratio test showing $\alpha(n+2) > \alpha(n)$ for $n < 5$ and $\alpha(n+2) < \alpha(n)$ for $n \geq 5$; (ii) strict concavity of $\log \alpha$ as a function of continuous dimension, guaranteeing uniqueness; and (iii) asymptotic decay establishing that no large- n competitor exists. This result elevates $n_C = 5$ from an input parameter to a derived consequence of extremal coupling, reducing BST's input count to zero. The entire Standard Model follows from a single variational principle: *among odd-dimensional type IV substrates, the universe selects the one that maximizes electromagnetic self-coupling.*

1. The Wyler-BST Formula

BST derives physics from the bounded symmetric domain

$$D_{IV}^{n_C} = \text{SO}_0(n_C, 2) / [\text{SO}(n_C) \times \text{SO}(2)]$$

of complex dimension n_C . The requirement that the color group $\text{SU}(N_c)$ act faithfully on the domain forces n_C to be odd, with $N_c = (n_C + 1)/2$. (Even n_C yields half-integer N_c , which is not a valid SU rank.)

The fine structure constant is determined by the Bergman kernel normalization of the S^1 fiber:

$$\alpha(n) = \frac{N_c^2}{8\pi^4} \left(\frac{\pi^n}{n! \cdot 2^{n-1}} \right)^{1/4} = \frac{1}{8\pi^4} \left(\frac{n+1}{2} \right)^2 \left(\frac{\pi^n}{n! \cdot 2^{n-1}} \right)^{1/4}$$

where n ranges over positive odd integers $\{1, 3, 5, 7, 9, \dots\}$.

Origin of each factor.

- $N_c^2 = ((n+1)/2)^2$: the Casimir of the fundamental representation of $\text{SU}(N_c)$, measuring the strength of the color sector's coupling to the S^1 electromagnetic fiber.
 - $\pi^n / (n! \cdot 2^{n-1})$: the Hua-normalized Bergman volume of D_{IV}^n , encoding the geometric capacity of the bulk.
 - $8\pi^4$: normalization from the Shilov boundary $\check{S} = S^{n_C-1} \times S^1$ and the fourth-root Weyl integral.
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2. Numerical Survey

We define $f(n) = N_c^2 \cdot (\pi^n / (n! \cdot 2^{n-1}))^{1/4}$ so that $\alpha(n) = f(n)/(8\pi^4)$. Evaluating at odd n :

n	N_c	$\alpha(n)$	$1/\alpha(n)$	Status
1	1	0.001708	585.3	
3	2	0.005472	182.7	
5	3	0.007297	137.04	MAXIMUM
7	4	0.006387	156.6	
9	5	0.004294	232.9	
11	6	0.002393	417.9	
13	7	0.001155	865.8	
15	8	0.000497	2013.4	

The maximum occurs at $n = 5$, yielding

$$\alpha(5) = \frac{9}{8\pi^4} \left(\frac{\pi^5}{1920} \right)^{1/4} = 0.00729735 \dots = 1/137.036 \dots$$

in agreement with the observed value to 0.6 ppm.

3. Ratio Test: The Core Proof

Theorem 1. *For odd $n \geq 1$, define $R(n) = f(n+2)/f(n)$. Then $R(n) > 1$ for $n \in \{1, 3\}$ and $R(n) < 1$ for all odd $n \geq 5$. Consequently, $\alpha(n)$ is strictly maximized at $n = 5$.*

Proof. We compute the ratio explicitly. Since

$$f(n) = \left(\frac{n+1}{2} \right)^2 \left(\frac{\pi^n}{n! \cdot 2^{n-1}} \right)^{1/4}$$

we have

$$R(n) = \frac{f(n+2)}{f(n)} = \underbrace{\left(\frac{n+3}{n+1} \right)^2}_{R_{\text{Casimir}}(n)} \times \underbrace{\left(\frac{\pi^2}{4(n+1)(n+2)} \right)^{1/4}}_{R_{\text{Vol}}(n)}$$

The first factor R_{Casimir} comes from the ratio of N_c^2 values:

$$\frac{((n+3)/2)^2}{((n+1)/2)^2} = \left(\frac{n+3}{n+1} \right)^2$$

The second factor R_{Vol} comes from the ratio of volume terms:

$$\frac{(\pi^{n+2}/((n+2)! \cdot 2^{n+1}))^{1/4}}{(\pi^n/(n! \cdot 2^{n-1}))^{1/4}} = \left(\frac{\pi^2}{(n+1)(n+2) \cdot 4} \right)^{1/4}$$

The competition. As n increases:

- $R_{\text{Casimir}}(n) = ((n+3)/(n+1))^2$ is a *decreasing* function that approaches 1 from above. It represents the growth of color sector coupling with dimension.
- $R_{\text{Vol}}(n) = (\pi^2/(4(n+1)(n+2)))^{1/4}$ is a *decreasing* function that approaches 0. It represents the dilution of Bergman volume in higher dimensions (the factorial in the denominator overwhelms the power of π).

The Casimir factor always exceeds 1 — adding dimensions always strengthens color coupling. The volume factor always falls below 1 for $n \geq 1$ (since $4(n+1)(n+2) > \pi^2$ already at $n = 1$). The question is which factor wins.

Evaluation at the critical values:

At $n = 3$:

$$R(3) = \left(\frac{6}{4}\right)^2 \times \left(\frac{\pi^2}{80}\right)^{1/4} = \frac{9}{4} \times 0.5927 = 1.3335 > 1$$

So $\alpha(5) > \alpha(3)$. $\square$

At $n = 5$:

$$R(5) = \left(\frac{8}{6}\right)^2 \times \left(\frac{\pi^2}{168}\right)^{1/4} = \frac{16}{9} \times 0.4923 = 0.8752 < 1$$

So $\alpha(7) < \alpha(5)$. $\square$

The crossover occurs between $n = 3$ and $n = 5$. The condition $R(n) = 1$ requires

$$\left(\frac{n+3}{n+1}\right)^8 = \frac{4(n+1)(n+2)}{\pi^2}$$

At $n = 3$: LHS = $(3/2)^8 = 25.63$, RHS = $80/\pi^2 = 8.11$. LHS > RHS, so $R(3) > 1$.

At $n = 5$: LHS = $(4/3)^8 = 9.99$, RHS = $168/\pi^2 = 17.02$. LHS < RHS, so $R(5) < 1$.

The crossover is clean: the LHS is rapidly decreasing while the RHS is rapidly increasing, so they cross exactly once.

4. Monotonic Decrease for $n \geq 5$

Theorem 2. $R(n)$ is strictly decreasing for $n \geq 1$, and $R(n) < 1$ for all $n \geq 5$. Consequently, $\alpha(n)$ is strictly decreasing for all odd $n \geq 5$.

Proof. We show $R(n+2) < R(n)$ for all $n \geq 1$. Define

$$R(n) = \left(\frac{n+3}{n+1}\right)^2 \left(\frac{\pi^2}{4(n+1)(n+2)}\right)^{1/4}$$

The first factor satisfies

$$\frac{R_{\text{Casimir}}(n+2)}{R_{\text{Casimir}}(n)} = \frac{((n+5)(n+1))^2}{((n+3)^2)^2} = \left(\frac{n^2+6n+5}{n^2+6n+9}\right)^2 < 1$$

since $n^2 + 6n + 5 < n^2 + 6n + 9$ for all n .

The second factor satisfies

$$\frac{R_{\text{Vol}}(n+2)}{R_{\text{Vol}}(n)} = \left(\frac{(n+1)(n+2)}{(n+3)(n+4)}\right)^{1/4} < 1$$

since $(n+1)(n+2) < (n+3)(n+4)$ for all $n \geq 0$.

Both sub-ratios are strictly less than 1, so their product is strictly less than 1. Therefore $R(n)$ is strictly decreasing.

Since $R(5) = 0.8752 < 1$ and R is decreasing, we have $R(n) < 1$ for all $n \geq 5$.

Explicitly:

n	$R(n)$
1	3.203
3	1.333
5	0.875
7	0.672
9	0.557
11	0.483
13	0.430
15	0.391

□

5. Concavity and Uniqueness (Continuous Extension)

For additional rigor, we extend α to a continuous function of a real variable $x > 0$ by replacing $n!$ with $\Gamma(x+1)$:

$$L(x) = \log \alpha(x) = 2 \ln\left(\frac{x+1}{2}\right) + \frac{1}{4} [x \ln \pi - \ln \Gamma(x+1) - (x-1) \ln 2] - \ln(8\pi^4)$$

Differentiating:

$$L'(x) = \frac{2}{x+1} + \frac{1}{4} [\ln \pi - \psi(x+1) - \ln 2]$$

$$L''(x) = -\frac{2}{(x+1)^2} - \frac{1}{4} \psi_1(x+1)$$

where $\psi = \Gamma'/\Gamma$ is the digamma function and ψ_1 is the trigamma function.

Both terms of L'' are strictly negative for all $x > 0$:

- $-2/(x+1)^2 < 0$ trivially.
- $-\psi_1(x+1)/4 < 0$ because the trigamma function is strictly positive for all positive arguments.

Therefore $L(x)$ is strictly concave on $(0, \infty)$.

A strictly concave function on an interval has at most one critical point, which must be a global maximum. Setting $L'(x^*) = 0$ and solving numerically gives

$$x^* = 5.200 \dots$$

The nearest odd integer to $x^* = 5.200$ is $n = 5$. Since L is concave and x^* lies between 5 and 7, the maximum over odd integers is at $n = 5$ (because $L(5) > L(3)$ by the ratio test, and $L(5) > L(7)$ by the ratio test, and concavity prevents any other odd integer from exceeding $L(5)$).

Remark. The continuous maximum at $x^* \approx 5.2$ is remarkably close to the odd integer 5. Nature does not have to “round far” — the extremal principle and the odd-integer constraint are nearly coincident.

6. Asymptotic Decay

Theorem 3. $\alpha(n) \rightarrow 0$ super-exponentially as $n \rightarrow \infty$.

Proof. By Stirling's approximation, $n! \sim \sqrt{2\pi n}(n/e)^n$, so

$$\frac{\pi^n}{n! \cdot 2^{n-1}} \sim \frac{\pi^n}{(n/e)^n \cdot 2^n} = \left(\frac{e\pi}{2n}\right)^n$$

For $n \geq 5$, we have $e\pi/(2n) < 1$, so the volume factor decays exponentially. Its fourth root decays as $\sim (e\pi/(2n))^{n/4}$, which overwhelms the polynomial growth of $N_c^2 = ((n+1)/2)^2$. Therefore $\alpha(n) \rightarrow 0$. $\square$

This rules out any “second maximum” at large n and confirms that $n = 5$ is the unique global maximum.

7. Summary of the Proof

Combining Theorems 1, 2, and 3:

1. $\alpha(1) < \alpha(3) < \alpha(5)$ — established by $R(1) > 1$ and $R(3) > 1$.
2. $\alpha(5) > \alpha(7) > \alpha(9) > \dots$ — established by $R(n) < 1$ for all $n \geq 5$ and $R(n)$ strictly decreasing.
3. $\alpha(n) \rightarrow 0$ — no large- n competitor exists.
4. Strict concavity of $\log \alpha$ guarantees uniqueness of the maximum.

Therefore $n = 5$ is the unique global maximizer of $\alpha(n)$ over all odd positive integers. $\blacksquare$

8. Physical Interpretation: Maximum Self-Coupling

8.1 What α Measures in BST

In BST, the fine structure constant α is not merely the electromagnetic coupling. It is the code rate of the substrate channel: the fraction of geometric capacity that manifests as observable interaction. Every photon exchange is a message sent through the S^1 fiber of D_{IV}^n , and α measures how efficiently that fiber communicates.

Two competing effects determine $\alpha(n)$:

- Color richness (N_c^2): more colors means more channels through which the S^1 fiber can couple. This grows with n .
- Geometric dilution (Bergman volume^{1/4}): higher-dimensional domains spread the coupling over exponentially more degrees of freedom. This shrinks with n .

At $n = 5$, these two effects are optimally balanced. Below $n = 5$, the substrate has too few color channels. Above $n = 5$, the geometric dilution overwhelms the color richness.

8.2 The Two Factors, Physically

$$\alpha(n) = \underbrace{\frac{N_c^2}{8\pi^4}}_{\text{color coupling}} \times \underbrace{\text{Vol}(D_{IV}^n)^{1/4}}_{\text{geometric capacity}}$$

- The color coupling wants n large (more $\text{SU}(N_c)$ generators, more interaction channels).
- The geometric capacity wants n small (less volume dilution, more concentrated coupling).

The maximum at $n = 5$ is the Nash equilibrium of these competing drives.

9. Shannon Capacity Interpretation

In BST's information-theoretic framework, the substrate is a noisy channel. Shannon's channel coding theorem states that reliable communication is possible if and only if the code rate does not exceed the channel capacity.

For a type IV substrate of dimension n :

- The channel is the S^1 electromagnetic fiber.
- The noise is the quantum fluctuation inherent in the Bergman kernel.
- The code rate is $\alpha(n)$.

Shannon's theorem demands operation at or near the capacity. Among odd-dimensional substrates, the maximum achievable code rate is $\alpha(5) = 1/137.036$. Any other dimension yields a strictly lower rate, meaning less efficient self-communication and slower dynamics.

The universe operates at Shannon capacity — it selects the substrate dimension that maximizes the rate of information exchange through its electromagnetic channel.

10. The Self-Starting Argument

At the moment of first commitment (the Big Bang), there is no external agent to select a dimension. The substrate must select itself. Consider the competition among odd-dimensional candidates:

1. Each candidate D_{IV}^n has a self-coupling strength $\alpha(n)$.
2. The rate of self-activation scales with α : stronger coupling means faster commitment (more rapid phase transition from uncommitted to committed substrate).
3. The dimension with the largest α activates first.
4. Once activated, it saturates the available degrees of freedom, precluding other dimensions.

This is a dynamical selection principle, analogous to:

- Spontaneous symmetry breaking (the field rolls to the deepest minimum).
- Competitive exclusion in ecology (the fastest-reproducing species dominates).
- Lasing threshold in quantum optics (the mode with highest gain activates first).

The “potential” governing this selection is $V(n) = -\alpha(n)$, and $n = 5$ is the unique global minimum.

11. Anthropic vs. Extremal Selection

It is essential to distinguish this result from anthropic reasoning:

	Anthropic	Extremal (BST)
Logic	“We observe $n = 5$ because only $n = 5$ permits observers”	“ $n = 5$ is selected because it maximizes $\alpha(n)$ ”
Requires observers?	Yes	No
Requires multi-universe?	Yes (other n exist but are unobserved)	No (other n never activate)
Predictive?	Weakly (post-dicts what we see)	Strongly (uniquely determines n)
Falsifiable?	Difficult	Yes: if α were not maximal at $n = 5$, BST would be wrong

The maximum- α principle is a dynamical selection, not an observational selection. It operates like a ball rolling to the bottom of a potential well. No observer is needed; no ensemble of universes is invoked.

12. The Zero-Input Claim

12.1 Before This Result

BST derived all Standard Model parameters from a single input: the complex dimension $n_C = 5$ of the type IV domain D_{IV}^5 . From $n_C = 5$ follows:

- $N_c = (n_C + 1)/2 = 3$ (three colors)
- $\alpha = (9/8\pi^4)(\pi^5/1920)^{1/4} = 1/137.036$ (fine structure constant)
- $N_{\max} = \lfloor 1/\alpha \rfloor = 137$ (maximum quantum number)
- $6\pi^5 m_e = 938.272$ MeV (proton mass from Yang-Mills gap)
- All mixing angles, mass ratios, coupling constants (see table below)

12.2 After This Result

The maximum- α principle derives $n_C = 5$ itself:

$$n_C = \arg \max_{n \text{ odd}} \alpha(n) = 5$$

The logical chain is now:

$$(\text{variational principle}) \longrightarrow n_C = 5 \longrightarrow \text{all of physics}$$

BST has zero free parameters.

The variational principle — “the substrate selects the odd dimension that maximizes its electromagnetic self-coupling” — is not an input in the usual sense. It is a *selection rule*, analogous to the principle of least action. Just as classical mechanics has zero inputs beyond the Lagrangian (and the principle of stationary action selects the trajectory), BST has zero inputs beyond the geometry (and the maximum- α principle selects the dimension).

12.3 The Complete Derivation Chain

$$\boxed{\max_{n \text{ odd}} \alpha(n)} \longrightarrow n_C = 5 \longrightarrow N_c = 3 \longrightarrow \alpha = \frac{1}{137.036}$$

$$\longrightarrow N_{\max} = 137 \longrightarrow m_p/m_e = 6\pi^5 \longrightarrow \text{everything}$$

13. Derived Quantities (Selected)

All from $n_C = 5$, which is derived from $\max \alpha$:

Quantity	BST Formula	Value	Precision
α	$(9/8\pi^4)(\pi^5/1920)^{1/4}$	1/137.036	0.0001%
m_p/m_e	$6\pi^5$	1836.15	0.002%
m_μ/m_e	$(24/\pi^2)^6$	206.769	0.003%

Quantity	BST Formula	Value	Precision
$\sin^2 \theta_W$	$N_c/(N_c + 2n_C) = 3/13$	0.2308	0.2% — Structural/Identified: unforced, MS-bar(M_Z), runs (K1263)
$\alpha_s(m_p)$	$(n_C + 2)/(4n_C) = 7/20$	0.350	—
η (baryon asymmetry)	$2\alpha^4/(3\pi)$	6.0×10^{-10}	1.4%
n_s (spectral index)	$1 - n_C/N_{\max} = 1 - 5/137$	0.9635	0.3σ
$\sin \theta_C$ (Cabibbo)	$1/(2\sqrt{n_C}) = 1/(2\sqrt{5})$	0.2236	0.3%
v (Fermi scale)	$m_p^2/(7m_e)$	246.01 GeV	0.046%
m_H (Higgs)	$(\pi/2)(1 - \alpha)m_W$	125.33 GeV	0.07%

14. Why This Was Not Obvious

Several features of the proof deserve emphasis:

14.1 The maximum is not at $n = 1$. Naively, one might expect the smallest dimension to win (least dilution). But the N_c^2 factor is too small at $n = 1$: a single color provides too little coupling.

14.2 The maximum is not at $n = 3$. The popular intuition that “three dimensions are special” does not apply here. At $n = 3$, $N_c = 2$, and $SU(2)$ does not provide enough color richness. The ratio $R(3) = 1.333$ shows that the step from $n = 3$ to $n = 5$ still gains more from color growth than it loses to dilution.

14.3 The crossover is sharp. $R(3) = 1.333 > 1$ and $R(5) = 0.875 < 1$. There is no “marginal” case. The maximum at $n = 5$ is robust: it is not a near-tie that could be upset by small corrections.

14.4 The continuous maximum is close to 5. The continuous extremum $x^* = 5.200$ falls between 5 and 7, decisively favoring $n = 5$ over $n = 7$. If x^* had fallen at, say, 5.9, one might worry about corrections tilting the balance. At 5.2, the margin is comfortable.

14.5 Strict concavity eliminates pathologies. Because $L''(x) < 0$ everywhere, there cannot be multiple local maxima, saddle points, or secondary peaks at large n . The maximum is unique and global.

15. The Structural Miracle

Consider what has been established:

1. The formula $\alpha(n)$ is determined by Bergman kernel geometry and $SU(N_c)$ representation theory. No free functions or adjustable parameters appear.
2. Among all odd positive integers, there is a unique maximum at $n = 5$.
3. At this maximum, the numerical value $\alpha(5) = 1/137.036$ matches experiment to better than 1 ppm.
4. The integer $N_{\max} = \lfloor 1/\alpha(5) \rfloor = 137$ is prime, which is essential for the discreteness of the quantum number spectrum.
5. From this single value, 40+ Standard Model parameters are derived to sub-percent accuracy.

The probability that all of this is coincidence is effectively zero. The maximum- α principle is either a profound truth about the structure of reality, or the most elaborate numerical coincidence in the history of physics.

16. Relation to Wyler (1969)

Armand Wyler first wrote down the formula for α in terms of D_{IV}^5 in 1969, deriving $\alpha = (9/8\pi^4)(\pi^5/1920)^{1/4}$. His result was criticized (notably by Robertson) for lacking a selection principle: *why D_{IV}^5 and not some other domain?*

The maximum- α principle answers Robertson's objection directly. Among all type IV domains of odd dimension, D_{IV}^5 is selected by a variational principle. Wyler had the right formula but not the right reason. BST provides both.

17. Conclusion

We have proved:

Theorem (Maximum-Alpha Selection). *The fine structure constant $\alpha(n)$, defined by the Wyler-BST formula over odd-dimensional type IV bounded symmetric domains, achieves its unique global maximum at $n = 5$. The proof is constructive (ratio test), confirmed by concavity (strict negativity of L''), and secured by asymptotic decay (Stirling bound).*

Corollary. *BST has zero input parameters. The complex dimension $n_C = 5$ is derived, not assumed, and all Standard Model constants follow.*

The maximum- α principle — *the substrate selects the geometry that maximizes its self-coupling* — is the single organizing idea from which all of known physics descends.

Appendix A: Complete Numerical Verification

n	N_c	alpha(n)	1/alpha(n)	R(n) = f(n+2)/f(n)
1	1	0.00170843	585.33	3.2032 (> 1)
3	2	0.00547243	182.73	1.3335 (> 1)
5	3	0.00729735	137.04	0.8752 (< 1) <-- MAX
7	4	0.00638690	156.57	0.6723 (< 1)
9	5	0.00429375	232.90	0.5573 (< 1)
11	6	0.00239283	417.92	0.4827 (< 1)
13	7	0.00115500	865.80	0.4300 (< 1)
15	8	0.00049668	2013.39	0.3906 (< 1)
17	9	0.00019400	5154.71	---

Appendix B: Ratio Decomposition

$$R(n) = \underbrace{\left(\frac{n+3}{n+1}\right)^2}_{R_C} \times \underbrace{\left(\frac{\pi^2}{4(n+1)(n+2)}\right)^{1/4}}_{R_V}$$

n	R_C	R_V	$R(n) = R_C \times R_V$
1	4.0000	0.8008	3.2032
3	2.2500	0.5927	1.3335
5	1.7778	0.4923	0.8752
7	1.5625	0.4303	0.6723
9	1.4400	0.3870	0.5573

At $n = 3 \rightarrow 5$: Casimir growth (2.25 $\times$) still outweighs volume dilution ($\times 0.59$), so α increases.

At $n = 5 \rightarrow 7$: Casimir growth (1.78 $\times$) can no longer compensate volume dilution ($\times 0.49$), so α decreases.

The crossover is governed by the exact condition

$$\left(\frac{n+3}{n+1}\right)^8 \stackrel{?}{>} \frac{4(n+1)(n+2)}{\pi^2}$$

which holds for $n \leq 3$ and fails for $n \geq 5$. ■